

Week 2: Ecosystem

JSS3 Intermediate Science — First Term Lesson Note (Deep-Dive Edition)

Introduction

This week introduces the concept of an ecosystem — how living things interact with each other and their environment, and how energy flows through these interactions.

Meaning and Components

An **ecosystem** is a community of living organisms interacting with each other and with their non-living environment. It has two main components:

Component	Description	Examples
Biotic factors	The living parts of an ecosystem	Plants, animals, microorganisms
Abiotic factors	The non-living parts of an ecosystem	Sunlight, water, soil, temperature, air

Interdependence and Balance of Nature

Living organisms depend on each other and on their environment for survival — this is called **interdependence**. For example, plants depend on soil nutrients and sunlight, herbivores depend on plants for food, and predators depend on other animals. This web of dependence keeps an ecosystem in **balance**; when one part is disturbed, it can affect the whole system.

Food Chains, Food Webs, and Energy Flow

A **food chain** shows the transfer of energy from one organism to another in a straight line, e.g.: **Grass → Grasshopper → Frog → Snake → Hawk**. A **food web** is a more realistic network showing many interconnected food chains in an ecosystem.

Trophic Level	Role	Example
Producers	Make their own food through photosynthesis	Green plants
Primary consumers	Herbivores that eat producers	Grasshopper, goat
Secondary consumers	Carnivores that eat primary consumers	Frog, lizard
Tertiary consumers	Top predators that eat secondary consumers	Hawk, snake
Decomposers	Break down dead organisms, returning nutrients to soil	Bacteria, fungi

Disruption of Ecosystems

Ecosystems can be disrupted by deforestation, pollution, overhunting, climate change, and the introduction of invasive species. Disruption at any trophic level can affect the entire food web,

sometimes leading to species decline or extinction.

Real-Life Application

Nigeria's diverse ecosystems — from the tropical rainforest in the south to the savanna grasslands in the north, and the mangrove swamps of the Niger Delta — each support unique food webs. Deforestation for farmland or logging in these areas can disrupt local food chains and reduce biodiversity.

Vocabulary

Word	Meaning
Biotic	Relating to living organisms
Abiotic	Relating to non-living components of an environment
Trophic level	A position in a food chain based on how an organism obtains energy

Evaluation Questions

- 1 Define an ecosystem.
- 2 Distinguish between biotic and abiotic factors, with two examples of each.
- 3 Draw or describe a simple food chain with four organisms.
- 4 Explain the difference between a food chain and a food web.
- 5 What is the role of decomposers in an ecosystem?
- 6 Name the four trophic levels of consumers, in order.
- 7 Explain what “interdependence” means in an ecosystem.
- 8 List three causes of ecosystem disruption.
- 9 Give one example of a Nigerian ecosystem and describe one organism found there.
- 10 In your own words, explain why disrupting one part of a food web can affect the whole ecosystem.